

SIMATS ENGINEERING

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Machine Learning

Experiment -7

Write a program to implement Logistic Regression (LR) algorithm in python

Code:

```
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score, confusion_matrix,
classification_report

# Load dataset
df = pd.read_csv("play_tennis (1).csv")

# Separate features and target
X = df.drop("play", axis=1)
y = df["play"]

# Convert categorical columns into numerical values if present
X = pd.get_dummies(X)

# Split dataset into training and testing sets
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

# Feature scaling
scaler = StandardScaler()
X_train = scaler.fit_transform(X_train)
X_test = scaler.transform(X_test)

# Create Logistic Regression model
model = LogisticRegression()

# Train the model
model.fit(X_train, y_train)

# Make predictions
y_pred = model.predict(X_test)

# Calculate accuracy
accuracy = accuracy_score(y_test, y_pred)

print("Logistic Regression")
print("-----")
print("Accuracy: {:.2f}%".format(accuracy * 100))

# Display confusion matrix
print("\nConfusion Matrix:")
print(confusion_matrix(y_test, y_pred))

# Display classification report
```

```

print("\nClassification Report:")
print(classification_report(y_test, y_pred))

# Display actual and predicted values
result = pd.DataFrame({
    "Actual": y_test.values,
    "Predicted": y_pred
})

print("\nActual vs Predicted:")
print(result)

```

OUTPUT:

```

... Logistic Regression
-----
Accuracy: 66.67%

Confusion Matrix:
[[0 1]
 [0 2]]

Classification Report:

```

	precision	recall	f1-score	support
No	0.00	0.00	0.00	1
Yes	0.67	1.00	0.80	2
accuracy			0.67	3
macro avg	0.33	0.50	0.40	3
weighted avg	0.44	0.67	0.53	3

```


Actual vs Predicted:
Actual Predicted
0    Yes      Yes
1    Yes      Yes
2    No       Yes

```